Exam Preparation

Introduction to Optimization

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The General Picture

Optimization is about tools, not specific problems.

Unconstrained / Simple Constraints

$$\boxed{\min_x f(x) \text{ subject to } x \in C}$$

- Rewrite as: $\min_x f(x) + \iota_C(x)$.
- Then immediately $0 \in \partial(f(x) + \iota_C(x))$.
- For closed convex C, $\partial\iota_C = N_C(x)$.
- If f smooth $\Rightarrow \partial f(x) = \nabla f(x)$.

Lagrangian / KKT / Duality

$$\boxed{\min_x f(x) \text{ subject to } g_i(x) \leq 0, h_j(x) = 0}$$

Define the Lagrangian

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \textstyle\sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x).$$

KKT conditions (use when f, g convex):

- Stationarity: $\nabla_x L = 0$.
- Primal feasibility: x^* satisfies all constraints.
- Dual feasibility: $\lambda_i \geq 0$.
- Complimentarity: $\lambda_i g_i(x^*) = 0$.

Subgradients, Conjugates, and Proximity

Conjugates

- Given f, $f^*(y) = \sup_x \{y^\top x - f(x)\}$ (closed & convex).
- Reciprocity: $y \in \partial f(x) \Leftrightarrow x \in \partial f^*(y)$.
- f is $\mu-$strongly convex $\Leftrightarrow f^*$ is $\frac{1}{\mu}-$smooth.
- Dual: $\min_x f(x) + g(Ax) \Rightarrow \min_y f^*(-A^\top y) + g^*(y)$.

Proximity Operator

- Defined as $y = \mathrm{prox}_f(x) = \arg\min_z \{f(z) + \frac{1}{2}\|x - z\|^2\}$.
- y satisfies $x - y \in \partial f(y)$ and $\mathrm{prox}_f = (I + \partial f)^{-1}$.
- prox_f behaves like a gradient step for nonsmooth f.
- If $f = \iota_C$, then $\mathrm{prox}_f = \mathrm{proj}_C$.

> **Pattern Recognition 1**
>
> When an algorithm update looks like $x^{k+1} = \mathrm{prox}_{\tau f}(x^k - \tau \nabla g(x^k))$, it's a **proximal gradient** or **primal-dual** step. Recognize that the prox handles nonsmooth parts.

> **Pattern Recognition 2**
>
> If the prox involves f^* or $A^\top y$, you're seeing a **dual update** (ADMM / Chambolle-Pock), then apply the **Moreau decomposition**: $\mathrm{prox}_{\lambda f}(x) + \lambda \mathrm{prox}_{f^* \lambda}(x\lambda) = x$.

2023 Exam

- **(Q1)** If φ is μ-strongly convex, then φ^* is $\left(\frac{1}{\mu}\right)$-smooth.
- **(Q2-3)** Assume $f : \mathbb{R}^N \to \mathbb{R}$ continuous, μ-strongly convex and let $V = \{x : Ax = b\}$. If minimizing f over V, show uniqueness of the solution and state the optimality condition.
- **(Q4-10)** Define the Lagrangian $\mathcal{L}(x, y) = f(x) + y^\top(Ax - b)$ and iterate

$$x^{k+1} = \arg\min_x \mathcal{L}(x, y^k),$$
$$y^{k+1} = y^k + \alpha\left(Ax^{k+1} - b\right).$$

  - Prove well-posedness,
  - derive the dual problem h,
  - compute ∇h,
  - show the y-update is gradient descent on h,
  - give stepsize $\alpha < \frac{2\mu}{\|A\|^2}$, and discuss rates.

(Q1) If φ is μ-strongly convex, then φ^* is $\left(\frac{1}{\mu}\right)$-smooth.

Facts:

- Reciprocity: $y \in \partial f(x) \Leftrightarrow x \in \partial f^*(y)$.
- Strong convexity: $(u-v)^\top(y-x) \geq \mu\|x-y\|^2$ whenever $u \in \partial f(y), v \in \partial f(x)$.

Proof

For $i = 1, 2$, let $z_i^* \in \partial\varphi(z_i)$. Then, by reciprocity, $z_i \in \partial\varphi^*(z_i^*)$. By strong convexity, $\mu\|z_1 - z_2\|^2 \leq (z_1^* - z_2^*)^\top(z_1 - z_2)$, and applying Cauchy Schwarz gives $\mu\|z_1 - z_2\|^2 \leq (z_1^* - z_2^*)^\top(z_1 - z_2) \leq \|z_1^* - z_2^*\|\|z_1 - z_2\|$. Thus, $\mu\|z_1 - z_2\| \leq \|z_1^* - z_2^*\|$, that is, $\|\nabla\varphi(z_1^*) - \nabla\varphi(z_2^*)\| \leq \frac{1}{\mu}\|z_1^* - z_2^*\|$.

Subdifferential of strongly convex is single valued

Assume z^* has two subgradients $x_1, x_2 \in \partial\varphi^*(z^*)$. By reciprocity, $z^* \in \partial\varphi(x_1)$ and $z^* \in \partial\varphi(x_2)$. From strong convexity with $u = v = z^*$, we get $0 = \langle z^* - z^*, x_1 - x_2\rangle \geq \mu\|x_1 - x_2\|^2$, hence $x_1 = x_2$.

(Q2-3) Assume $f : \mathbb{R}^N \to \mathbb{R}$ continuous, μ-strongly convex and let $V = \{x : Ax = b\}$. If minimizing f over V, show uniqueness of the solution and state the optimality condition.

(Q2) Uniqueness

- We immediately recognize this is of the form $f + \iota_V$.
- f is continuous, ι_V is closed, $\Rightarrow f + \iota_V$ is closed.
- f is strongly convex, ι_V is convex $\Rightarrow f + \iota_V$ is strongly-convex $\Rightarrow$ unique solution.

(Q3) Optimality Condition

- From Fermat's rule, $0 \in \partial f(x^*) + \partial \iota_V(x^*)$.
- Since $V = \{x : Ax = b\}$ is affine, $\partial \iota_V(x^*) = N_V(x^*) = \mathrm{ran}(A^\top)$.
- From the KKT conditions, there exists a y^* such that $-A^\top y^* \in \partial f(x^*)$ and $Ax^* = b$.

Tip: refresh your linear algebra (kernels, rank, matrix derivatives) and analysis knowledge (convergence, sequence, series).

(Q4) Prove $x^{k+1} = \arg\min_x \mathcal{L}(x, y^k) = \arg\min_x \{f(x) + y^\top (Ax - b)\}$ is well defined.

Proof

For fixed y^k, the function $x \to \mathcal{L}(x, y^k) = f(x) + \left(A^\top y^k\right)^\top x - \left(y^k\right)^\top b$ is the sum of a closed, μ-strongly convex function f and an affine function. Therefore it is closed and μ-strongly convex, hence admits a unique minimizer x^{k+1}.

(Q5) State the optimality condition for $x^{k+1} = \arg\min_x \mathcal{L}(x, y^k)$.

Proof

The optimality condition is $0 \in \partial_x \mathcal{L}\left(x^{k+1}, y^k\right)$. In other words, $0 \in \partial f\left(x^{k+1}\right) + A^\top y^k \Leftrightarrow -A^\top y^k \in \partial f\left(x^{k+1}\right)$.

(Q6) Find the dual problem of minimizing f over V.

We already saw the primal problem is $\min_x f(x) + \iota_V(x)$. Two equivalent routes:

Fenchel-Rockafellar Dual

- The general Fenchel-Rockafellar dual problem is

$$\min_y f^*(-A^\top y) + g^*(y).$$

- g^* for $g = \iota_{\{b\}}$ is

$$g^*(y) = \sup_z \left\{ y^\top z - \iota_{\{b\}}(z) \right\} = \sup_{\{z=b\}} y^\top z = b^\top y.$$

- The dual is then $\min_y h(y) = \min_y \{ f^*(-A^\top y) + b^\top y \}$.

Lagrangian Dual

- The Lagrangian is $\mathcal{L}(x, y) = f(x) + y^\top (Ax - b)$.
- The dual function is $d(y) = \inf_x \mathcal{L}(x, y) = -b^\top y + \inf_x \left\{ f(x) + (A^\top y)^\top x \right\}$.
- By definition of the conjugate, $\inf_x \{ f(x) + u^\top x \} = -\sup_x \{ (-u^\top x) - f(x) \} = -f^*(-u)$ with $u = A^\top y$.
- Then, $d(y) = -b^\top y - f^*(-A^\top y)$.
- The Lagrangian dual is $\max_y d(y)$, i.e., $\max_y \{ -b^\top y - f^*(-A^\top y) \} = \min_y \{ f^*(-A^\top y) + b^\top y \}$.

(Q7) Verify that the dual $h(y) = f^*(-A^\top y) + b^\top y$ is L-smooth with $L = \frac{\|A\|^2}{\mu}$.

Proof

Differentiating, we get $\nabla h(y) = -A \nabla f^*(-A^\top y) + b$. Recall that f^* is $\frac{1}{\mu}$-smooth. We apply the defintion of h, factor out A, the b's cancel out, and use the smoothness of f^*:

$$\|\nabla h(y_1) - \nabla h(y_2)\| \leq \|A\| \left\|\nabla f^*(-A^\top y_1) - \nabla f^*(-A^\top y_2)\right\| \leq \|A\| \frac{1}{\mu} \left\|A^\top (y_1 - y_2)\right\| \leq \frac{\|A\|^2}{\mu} \|y_1 - y_2\|.$$

(Q8) Show the sequence $y^{k+1} = y^k + \alpha\left(Ax^{k+1} - b\right)$ satisfies $y^{k+1} = y^k - \alpha \nabla h\left(y^k\right)$.

Proof

In **(Q5)**, we showed $-A^\top y^k \in \partial f\left(x^{k+1}\right)$, and by reciprocity, $x^{k+1} = \nabla f^*\left(-A^\top y^k\right)$.
Then, $\nabla h\left(y^k\right) = -A \nabla f^*\left(-A^\top y^k\right) + b = -Ax^{k+1} + b = -\left(Ax^{k+1} - b\right)$, hence $y^{k+1} = y^k - \alpha \nabla h\left(y^k\right)$.

(Q9) For which values of α can we guarantee convergence of (x^k, y^k) to an optimal pair?.

Answer

- In **(Q8)**, we showed that $y^{k+1} = y^k - \alpha \nabla h(y^k)$. So, we are doing **vanilla gradient descent** on h.
- Since h is convex and L-smooth with $L = \frac{\|A\|^2}{\mu}$, vanilla GD converges for $0 < \alpha \leq \frac{2}{L} = \frac{2\mu}{\|A\|^2}$.
- Since the primal problem has a unique minimizer, h has a minimizer y^* for which $\nabla h(y^*) = 0$.
- From **(Q8)**, $\nabla h(y^k) = -(Ax^{k+1} - b) = 0$ gives $Ax^* = b$ (feasibility).
- If $y = y^*$ is fixed, $x^{k+1} = \arg\min_x \mathcal{L}(x, y^*)$ does nothing after one iteration, hence $(x^k, y^k) \to (x^*, y^*)$.

(Q9) What is the convergence rate of this algorithm?.

Answer

- Vanilla GD for convex and L-smooth: $h\left(y^k\right) - h^* = \mathcal{O}\left(\frac{1}{k}\right)$.
- If h strongly convex (e.g, if f is smooth so that f^* is strongly convex): linear with $h\left(y^k\right) - h^* \leq \left(1 - \alpha\mu_h\right)^k \left(h\left(y^0\right) - h^*\right)$.

Takeaway & Pattern Recognition

- We started with a general optimization problem.
- We proved well-posedness and properties that come up later.
- We use the same tools over and over (KKT, optimality conditions, duality).
- From duality, we reduce the problem to something simpler (vanilla GD).

2024 Exam

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be CCP with a nonempty set of minimizers.
Goal: find the minimizer of f with the smallest norm.

Idea

- Not one way to solve, we have a **set of tools** to use.
- We will add a small penalty $\frac{\varepsilon}{2}\|x\|^2$.
- Why? To enforce strong convexity and bias the solution toward a small norm.
- Then, we will probably prove some convergence to show the two minimizers (original and penalized) are related.

For the following, let's define $f_\varepsilon : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ by $f_\varepsilon(x) = f(x) + \frac{\varepsilon}{2}\|x\|^2$.

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What is the first thing we can ask?

(Q1) Is there a solution, and is it unique?

Answer 1

Tool: closure and strong convexity preservation.

- f is proper, closed, convex.
- $\frac{\varepsilon}{2}\|x\|^2$ is ε-strongly convex.
- The sum is closed and ε-strongly convex $\Rightarrow$ unique solution, denoted x_ε.

Answer 2

Tool: subgradient & strong convexity definition.

- ∂f is monotone: $\langle u - v, x - y\rangle \geq 0$ for all $u \in \partial f(x), v \in \partial f(y)$ and $\partial f_\varepsilon(x) = \partial f(x) + \varepsilon x$.
- Let $u \in \partial f(x), v \in \partial f(y)$, so that $p = u + \varepsilon x \in \partial f_\varepsilon(x)$ and $q = v + \varepsilon y \in \partial f_\varepsilon(y)$.
- Then, $\langle p - q, x - y\rangle = \langle (u - v) + \varepsilon(x - y), x - y\rangle = \underbrace{(\langle u - v, x - y\rangle)}_{\geq 0 \text{ by monotonicity}} + \varepsilon\|x - y\|^2 \geq \varepsilon\|x - y\|^2$.

(Q2) Prove that $\min f + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \le f_\varepsilon(x_\varepsilon) \le f_\varepsilon(x)$ for all $x \in \mathbb{R}^n$.

Proof

- Take any $x \in \mathbb{R}^n$.
- **(A)** Since x_ε is the minimizer of f_ε, we have $f_\varepsilon(x_\varepsilon) \le f_\varepsilon(x)$.
- **(B)** We trivially have $\min(f) \le f(x)$, and similarly, $\min(f) \le f(x_\varepsilon)$.
- **(C)** Adding $\frac{\varepsilon}{2}\|x_\varepsilon\|^2$ to both sides of **(B)**, $\min f + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \le f(x_\varepsilon) + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 = f_\varepsilon(x_\varepsilon)$.
- Combining **(A)** and **(C)** gives $\min f + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \le f_\varepsilon(x_\varepsilon) \le f_\varepsilon(x)$.

Tool: optimality of x_ε and definition of f_ε.

(Q3) Prove that $\|x_\varepsilon\| \leq \|x^*\|$, where x^* is the minimizer of f with the smallest norm.

Proof

- From **(Q2)** we have $f_\varepsilon(x_\varepsilon) \leq f_\varepsilon(x)$ for all $x \in \mathbb{R}^n$. Expanding, $f(x_\varepsilon) + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq f(x) + \frac{\varepsilon}{2}\|x\|^2$.
- **(A)** Pick $x^* \in \arg\min f$, then $f(x_\varepsilon) + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq \min f + \frac{\varepsilon}{2}\|x^*\|^2$.
- **(B)** Trivially, we again have $\min f \leq f(x_\varepsilon)$.
- Combining **(A)** and **(B)**, we get $\min f + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq f(x_\varepsilon) + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq \min f + \frac{\varepsilon}{2}\|x^*\|^2$.
- Looking at the left and rightmost parts of the inequality, $\|x_\varepsilon\|^2 \leq \|x^*\|^2 \Rightarrow \|x_\varepsilon\| \leq \|x^*\|$.

Tool: optimality of x_ε, definition of f_ε, and sandwiching.

(Q4) Show that $\lim_{\varepsilon \to 0} f(x_\varepsilon) = \min f$.

Proof

- From the intermediate step in **(Q3)**, we had $\min f + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq f(x_\varepsilon) + \frac{\varepsilon}{2}\|x_\varepsilon\|^2 \leq \min f + \frac{\varepsilon}{2}\|x^*\|^2$.
- Dropping the $\frac{\varepsilon}{2}\|x_\varepsilon\|^2$ from the two parts on the left preserves the inequalities and gives $\min f \leq f(x_\varepsilon) \leq \min f + \frac{\varepsilon}{2}\|x^*\|^2$.
- Now, take $\varepsilon \to 0$, the RHS goes to $\min f$ and the LHS is already $\min f$.
- Hence, $\lim_{\varepsilon \to 0} f(x_\varepsilon) = \min f$.

(Q5) Show that $x_\varepsilon \to x^*$ as $\varepsilon \to 0$.

Proof

- The sequence (x_ε) is bounded since $\|x_\varepsilon\| \leq \|x^*\|$, hence has cluster points.
- Any cluster point $\overline{x}$ satisfies: $f(\overline{x}) = \min f$, since $f\left(x_{\varepsilon_k}\right) \to \min f$ and $f(\overline{x}) \leq \lim_{k \to \infty} \inf f\left(x_{\varepsilon_k}\right) = \min f$.
- By **(Q3)**, $\|x_\varepsilon\| \leq \|x^*\|$, so it has the minimal norm, hence $x_\varepsilon \to x^*$.

Let $\varepsilon_k \to 0$ be a positive real sequence and define a sequence (x^k) by iterating

$$x^{k+1} = \operatorname{prox}_{\gamma f_{\varepsilon_k}}(x_k) = \arg\min_x \left\{ f_{\varepsilon_k} + \frac{1}{2\gamma}\|x - x_k\|^2 \right\}.$$

(Q6) What is the optimality condition?

Answer

- By this point, easy. It is $0 \in \partial\left(f_{\varepsilon_k}(x^{k+1}) + \frac{1}{2}\gamma\|x^{k+1} - x^k\|^2 \right)$.
- Rewriting, we get $0 \in \partial f_{\varepsilon_k}(x^{k+1}) + \frac{1}{\gamma}(x^{k+1} - x^k)$.
- Alternatively, $-\frac{1}{\gamma}(x^{k+1} - x^k) \in \partial f_{\varepsilon_k}(x^{k+1})$.
- Let's define $g_k \in \partial f_{\varepsilon_k}(x^{k+1})$ as $g_k = -\frac{1}{\gamma}(x^{k+1} - x^k)$.

(Q7) Use the strong convexity f_ε to show

$$f_{\varepsilon_k}(x^*) \geq f_{\varepsilon_k}\left(x^{k+1}\right) - \frac{1}{\gamma}\left(x^{k+1} - x^k\right)\left(x^* - x^{k+1}\right) + \frac{\varepsilon_k}{2}\left\|x^{k+1} - x^*\right\|^2.$$

Proof

From the definition of strong convexity and **(Q6)** with $-\frac{1}{\gamma}\left(x^{k+1} - x^k\right) = g_k \in \partial f_{\varepsilon_k}\left(x^{k+1}\right)$, we get $f_{\varepsilon_k}(x^*) \geq f_{\varepsilon_k}\left(x^{k+1}\right) + g_k\left(x^* - x^{k+1}\right) + \frac{\varepsilon_k}{2}\left\|x^{k+1} - x^*\right\|^2$, which is exactly the statement above.

Warning: lots of algebra coming!

If skipping something, this should be it.

(Q8) Now, show the contractive property

$$(1+\gamma\varepsilon_k)\left\|x^{k+1}-x^*\right\|^2 \le \left\|x^k-x^*\right\|^2 + \gamma\varepsilon_k\left[\|x^*\|^2 - \left\|x_{\varepsilon_k}\right\|^2\right].$$

Proof

- Start from $f_{\varepsilon_k}(x^*) \ge f_{\varepsilon_k}(x^{k+1}) - \frac{1}{\gamma}(x^{k+1}-x^k)(x^*-x^{k+1}) + \frac{\varepsilon_k}{2}\left\|x^{k+1}-x^*\right\|^2$.
- Use the three-point identity: $-\langle x^{k+1}-x^k, x-x^{k+1}\rangle = \frac{1}{2}\left(\left\|x-x^k\right\|^2 - \left\|x-x^{k+1}\right\|^2 - \left\|x^{k+1}-x^k\right\|^2\right)$.
- Take $x = x^*$, substitute and rearrange:

  $(1+\gamma\varepsilon_k)\left\|x^*-x^{k+1}\right\|^2 \le \left\|x^*-x^k\right\|^2 - \left\|x^{k+1}-x^k\right\|^2 + 2\gamma\left(f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}(x^{k+1})\right)$.
- **(A)** Drop the non-positive term to get $(1+\gamma\varepsilon_k)\left\|x^*-x^{k+1}\right\|^2 \le \left\|x^*-x^k\right\|^2 + 2\gamma\left(f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}(x^{k+1})\right)$.
- **(B)** Since x_{ε_k} minimizes f_{ε_k}, we have $f_{\varepsilon_k}\left(x_{\varepsilon_k}\right) \le f_{\varepsilon_k}(x^{k+1})$ and $f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}(x^{k+1}) \le f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}\left(x_{\varepsilon_k}\right)$.
- From **(Q2)**, we have $\min f = f(x^*) \le f\left(x_{\varepsilon_k}\right)$.
- **(C)** Thus, $f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}\left(x_{\varepsilon_k}\right) = \left(\min f + \frac{\varepsilon_k}{2}\|x\|^2\right) - \left(f\left(x_{\varepsilon_k}\right) + \frac{\varepsilon_k}{2}\|\varepsilon_k\|^2\right) \le \frac{\varepsilon_k}{2}\left(\|x^*\|^2 - \left\|x_{\varepsilon_k}\right\|^2\right)$.
- **(D)** Combining **(B)** and **(C)**, we get $f_{\varepsilon_k}(x^*) - f_{\varepsilon_k}(x^{k+1}) \le \frac{\varepsilon_k}{2}\left(\|x^*\|^2 - \left\|x_{\varepsilon_k}\right\|^2\right)$.
- Plug **(D)** into **(A)** to get exactly $(1+\gamma\varepsilon_k)\left\|x^{k+1}-x^*\right\|^2 \le \left\|x^k-x^*\right\|^2 + \gamma\varepsilon_k\left[\|x^*\|^2 - \left\|x_{\varepsilon_k}\right\|^2\right]$.

(Q9) Use the Lemma below (which you do not need to prove) to show that if $\sum_{k\geq 0} \varepsilon_k = \infty$, then $x_k \to x^*$ as $k \to \infty$.

Summability Lemma

Let (A_k), (h_k), and (δ_k) be nonnegative real sequences such that $(1+\delta_k)A_{k+1} \leq A_k + \delta_k h_k$ for every $k \geq 0$. If $h_k \to 0$ as $k \to \infty$ and $\sum_{k\geq 0} \delta_k = \infty$, then $A_k \to 0$ as $k \to \infty$.

Proof

- Let $A_k = \left\|x^k - x_{\varepsilon_k}\right\|^2$, $h_k = \left\|x^{k+1} - x^k\right\|^2$, $\delta_k = \gamma\varepsilon_k$.
- Then, **(Q8)** is exactly $(1+\delta_k)A_{k+1} \leq A_k + \delta_k h_k$.
- Rearrange $(1+\delta_k)A_{k+1} \leq A_k + \delta_k h_k$ as $A_k - A_{k+1} \geq h_k + \delta_k A_{k+1} \geq h_k$.
- Summing $A_k - A_{k+1} \geq h_k$ gives $A_0 - A_N \geq \sum h_k$, so $\infty > A_0 \geq \sum h_k$, then $h_k \to 0$.
- If $\sum_{k\geq 0} \varepsilon_k = \infty$, we have $\sum \delta_k \to \infty$.
- Thus $A_k \to 0$, so $\left\|x_k - x_{\varepsilon_k}\right\| \to 0$.
- We already showed $x_{\varepsilon_k} \to x^*$ (least norm minimizer). Therefore, $x^k \to x^*$.

Takeaway (again)

Checklist

- We are given any problem.
- We show it is well-posed.
- Often many ways to solve it:
  - Direct methods (find explicitly).
  - Iterative methods (gradients for smooth / proximal for non-smooth).
- Fermat's rule, KKT, subdifferentials, duality: almost always applicable.
- Constraints:
  - Indicator function and normal cones (simple constraints)
  - Lagrangian + KKT (more complex constraints)

Cheat Sheet

Definitions

- (Strong) convexity, with subgradients
- Smoothness (Baillon-Haddad)
- First-order optimality, KKT conditions
- Combinations of (strongly) convex functions
- Duality, reciprocity, Fenchel-Young
- Normal cones for common sets (linear/affine subspaces)
- Matrix kernels, orthogonality, sequence convergence

Algorithms

- Convergence rates + definitions
- Convergence when convex / smooth/ strongly-convex
- Common update rules + step size criteria
- The proximity operator + properties
- Dual problems (Fenchel-Rockafellar / Lagrange)
- Linear programming (exact solutions)

Good luck on the exam!

Fill in the course evaluation!

Best, Alex.